

Making games is an ideal way to have fun while learning a language. It's deeply satisfying to play a game you wrote, and writing a simple game will teach you a lot about how professionals develop games. As you work through this chapter, enter and run the code to identify how each code block contributes to overall gameplay. Experiment with different values and settings to better understand how to refine interactions in your games.

NOTE

Alien Invasion spans a number of different files, so make a new `alien_invasion` folder on your system. Be sure to save all files for the project to this folder so your `import` statements will work correctly.

Also, if you feel comfortable using version control, you might want to use it for this project. If you haven't used version control before, see Appendix D for an overview.

Planning Your Project

When you're building a large project, it's important to prepare a plan before you begin to write code. Your plan will keep you focused and make it more likely that you'll complete the project.

Let's write a description of the general gameplay. Although the following description doesn't cover every detail of *Alien Invasion*, it provides a clear idea of how to start building the game:

In *Alien Invasion*, the player controls a rocket ship that appears at the bottom center of the screen. The player can move the ship right and left using the arrow keys and shoot bullets using the spacebar. When the game begins, a fleet of aliens fills the sky and moves across and down the screen. The player shoots and destroys the aliens. If the player destroys all the aliens, a new fleet appears that moves faster than the previous fleet. If any alien hits the player's ship or reaches the bottom of the screen, the player loses a ship. If the player loses three ships, the game ends.

For the first development phase, we'll make a ship that can move right and left when the player presses the arrow keys and fire bullets when the player presses the spacebar. After setting up this behavior, we can create the aliens and refine the gameplay.

Installing Pygame

Before you begin coding, install Pygame. We'll do this the same way we installed `pytest` in Chapter 11: with `pip`. If you skipped Chapter 11 or need a refresher on `pip`, see "Installing `pytest` with `pip`" on page 210.

To install Pygame, enter the following command at a terminal prompt:

```
$ python -m pip install --user pygame
```

If you use a command other than `python` to run programs or start a terminal session, such as `python3`, make sure you use that command instead.